The Mouse Race

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EE22005: Engineering Practice and Design

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22 March 2026

Executive Summary - individual

See the instructions within the lab script for writing an Executive Summary

Table of Contents

[Right click on table of contents and 'Update Field'.]

Executive Summary i
Table of Contents ii
List of Figures iii
List of Tables iv
List of Acronyms v
1. Introduction 1
1.1 Aims 1
1.2 Scope 1
2 Team management 1
2.1 Team leadership 1
3 Mouse Design 1
3.1 Dynamic system capture and simulation using SIMULINK 1
3.2 Power electronics simulation and design evaluation 1
3.3 Software design 2
4 Testing & Validation 2
4.1 Design & implementation of electronic circuit to vary motor speed 2
4.2 Test and validation of Code 2
5 Final Product 3
6 Investigation of complex problem with laboratory skills 3
7 Sustainability 3
7.1 Sustainable Energy Management 3
8 Risk and change management 3
8.1 Risk mitigation 3
8.2 Change management 3
9 Conclusion 3
10 Table, Figures and Equations examples [Delete] 4
References (including Quality Referencing) 5
Appendices 1
Appendix A: Design and Development Matrix 1
Appendix B: Risk Assessment & Mitigation 2

List of Figures

[Right click on list of figures and 'Update Field'. Delete page if there are no figures.]

Figure 1: Possible image of the racetrack 1
Figure 2: Sample Figure Caption 5

List of Tables

[Right click on list of tables and 'Update Field'. Delete page if there are no tables.]

Table 2 Sample Table Caption 3
Table 2 This is caption..... 3

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List of Acronyms

UoB	University of Bath

1. Introduction

1.1 Aims

1.2 Scope

2 Team management

2.1 Team Leadership

2.2 Being Security Minded - individual

A significant security consideration within this project is the protection of user energy consumption data and prevention of unauthorised access to the system. The system records time-dependent current data, which can indirectly reveal sensitive behavioural patterns such as occupancy, appliance usage, and daily routines. This makes data protection important, especially because embedded and IoT-style systems can introduce privacy and cybersecurity risks [3], [4].

Adopting a Security-Minded Approach and Responsible Judgement

A security-minded approach was adopted by recognising that even a small embedded monitoring system can generate sensitive data. Rather than assuming low risk, responsible judgement was applied by limiting the system to essential electrical measurements only. The design avoids collecting personal identifiers and prioritises local data storage rather than unnecessary cloud transmission. This follows the risk-based approach recommended in recognised cybersecurity frameworks [2].

Understanding and Complying with Security Codes

The project aligns with information-security principles such as controlled access, risk management, and protection of stored data. ISO/IEC 27001 is relevant because it provides a structured framework for managing information security risks in systems that collect, store, or process data [1]. Although this prototype is not a full commercial IoT product, applying these principles improves professionalism and user trust.

Ensuring Security-Minded Communication

Security-minded communication was considered when presenting graphs, logs, and CSV files. Results should only show necessary technical information and avoid exposing identifiable user routines. This reduces the risk of unintentionally sharing private behavioural data, which is particularly important in smart monitoring and IoT systems [3], [4].

Understanding Security Management Policies

Basic security management was applied through controlled file handling, local storage, and version control. If the project were expanded, stronger policies such as encrypted storage, authentication, and access permissions would be needed. This reflects the ISO/IEC 27001 principle that security should be managed throughout the system lifecycle rather than treated as a one-off task [1].

Contributing to Awareness of Security

The project contributes to security awareness by showing that energy data can reveal more than power consumption. By documenting this risk, the project encourages users and engineers to think about privacy when designing monitoring systems. This is important because cybersecurity guidance highlights awareness and risk identification as key parts of secure system design [2], [3].

Conclusion

This project demonstrates that security is important even in a small smart energy monitoring prototype. By adopting a security-minded approach, applying responsible judgement, following recognised security principles, managing data carefully, and increasing awareness, the system reduces the risk of data exposure while maintaining its energy-monitoring function.

3 Circuit Simulation

3.1 Circuit Diagram

3.2 Key Circuit Waveforms

3.3 Circuit Explanation

3.4 Simulation Relevance

4 Programming, Design, Testing and Validation

4.1 Design and Implementation of Python Code

This code presents a Python-based data analysis tool designed to process and evaluate time-series electrical measurements, specifically focusing on current readings from a device (e.g., a fan). The system reads raw data from a CSV file, performs statistical and operational analysis, and generates visualisations to support interpretation. The primary goal is to extract meaningful performance metrics such as power usage, energy consumption, and operational behaviour over time.

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Code Description

The program is structured using an object-oriented approach, with clear separation between data handling, analysis, and visualisation.

- **Data Importing:**
  The function `read_value_csv` reads time and current values from a CSV file and converts each row into a `ValueReading` object. This ensures the data is stored in a structured and accessible format.
- **Data Representation:**
  The `ValueReading` class encapsulates individual measurements, providing methods to retrieve time in different units (milliseconds, seconds, minutes) and the recorded value.
- **Data Analysis:**
  The `ValueAnalysis` class performs the core computations. It includes methods to calculate:
  - Root Mean Square (RMS) current for values above a threshold
  - Peak, mean, and standard deviation of the signal
  - Duty cycle based on ON/OFF periods
  - Transition detection between operational states
    The analysis assumes a threshold (300 units) to distinguish between active (ON) and inactive (OFF) states.
- **Energy Estimation:**
  The code estimates power and energy consumption using RMS current and a fixed power factor. It also calculates energy contribution over time for detailed analysis.
- **State Detection:**
  The program identifies ON and OFF intervals and logs transitions, enabling insight into device behaviour and runtime characteristics.
- **Visualisation:**
  Using `matplotlib`, the program generates multiple plots:
  - Current vs Time
  - Energy Consumption over Time
  - Device State (ON/OFF) over Time
    These visualisations help in understanding trends and operational patterns.
- **Summary Output:**
  A summary function prints key performance metrics, including runtime, RMS current, peak current, estimated energy consumption, and duty cycle.

```
import csv
import statistics
import matplotlib.pyplot as plt
import math
import numpy as np
```

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```python
# ---------IMPORTING---------
@staticmethod
def read_value_csv(filename):
    """
    Reads a CSV file containing time and value data.

    Args:
        filename (str): Path to the CSV file.

    Returns:
        List[ValueReading]: A list of ValueReading objects created from the
file.
    """
    readings = []

    with open(filename, mode='r', newline='') as file:
        reader = csv.reader(file)

        for row in reader:
            if not row:
                continue

            time = int(row[0])
            value = float(row[1])

            reading = ValueReading(time,value)
            readings.append(reading)

    return readings

# ---------VALUE MANIPULATION---------
class ValueReading:
    def __init__(self, time, value):
        """
    Represents a single measurement consisting of time and value.

    Attributes:
        time (int): Time of the reading in milliseconds.
        value (float): Measured value (e.g. current).
    """
        self.value = value
        self.time = time

    def GetValue(self):
        """Returns the recorded value."""
        return self.value
```

```
    def GetTime(self):
        """Returns the time in milliseconds."""
        return self.time

    def GetSeconds(self):
        """Returns the time converted to seconds."""
        return self.time / 1000

    def GetMinutes(self):
        """Returns the time converted to minutes."""
        return self.time / 60000

# ---------VALUE ANALYSIS---------
class ValueAnalysis:
    """
        Calculates RMS (Root Mean Square) of values above threshold (300).

        Returns:
            float: RMS value.
        """
    def __init__(self, lowertime, uppertime, file):
        self.ValueReadings = read_value_csv(file)
        self.lowertime = lowertime
        self.uppertime = uppertime

        self.BuildTimeline()

# ---------DATA ANALYSIS FEATURES---------
    def RMS(self):
        """
        Calculates RMS (Root Mean Square) of values above threshold (300).

        Returns:
            float: RMS value.
        """
        Values = [each_value.GetValue() for each_value in self.ValueReadings
if each_value.GetValue() > 300]
        squared_sum = sum(v ** 2 for v in Values)
        mean_squared = squared_sum / len(Values)
        return math.sqrt(mean_squared)


    def Peak(self):
        """
        Finds the maximum value in the dataset.
```

```
        Returns:
            float: Peak value.
        """
        Values = [each_value.GetValue() for each_value in self.ValueReadings]
        peak = Values[0]
        for each_value in Values:
            if each_value > peak:
                peak = each_value

        return round(peak,2)


    def StartUpTransientMag(self):
        """
        Placeholder for startup transient magnitude calculation.

        Returns:
            None
        """
        return None

    def Mean(self):
        """
        Calculates mean of values above threshold (300).

        Returns:
            float: Mean value.
        """
        Values = [each_value.GetValue() for each_value in self.ValueReadings
if each_value.GetValue() > 300]

        sum = 0
        num = 0
        for each_value in Values:
            sum += each_value
            num += 1

        return round(sum/num,2)


    def Stdev(self):
        """
        Calculates standard deviation of values above threshold.
```

```
        Returns:
            float: Standard deviation.
        """
        Values = [each_value.GetValue() for each_value in self.ValueReadings
if each_value.GetValue() > 300]
        return round(statistics.pstdev(Values),2)



    def DutyCycle(self):
        """
        Calculates duty cycle based on ON periods.

        Returns:
            float: Duty cycle percentage.
        """
        times = [each_time.GetTime() for each_time in self.ValueReadings if
each_time.GetValue() > 300]

        interval = 200

        on_periods = []

        start_on = times[0]

        for i in range(len(times)-1):
            diff = times[i+1] - times[i]

            if diff != interval:
                on_periods.append((start_on, times[i]))
                start_on = times[i+1]

        on_periods.append((start_on, times[-1]))

        total_on = 0
        for each_period in on_periods:
            total_on += each_period[1] - each_period[0]

        total_time = self.ValueReadings[-1].GetTime()

        return (total_on / total_time) * 100



# ---------STATE DETECTION---------
    def StateDetection(self, state):
        """
```

```
        Identifies ON or OFF periods.

        Args:
            state (str): "On" or "Off"

        Returns:
            str: Human-readable time ranges.
        """
        times = [each_time.GetTime() for each_time in self.ValueReadings if
each_time.GetValue() > 300]

        interval = 200

        on_periods = []
        off_periods = []

        start_on = times[0]

        for i in range(len(times)-1):
            diff = times[i+1] - times[i]

            if diff != interval:
                on_periods.append((start_on, times[i]))
                off_periods.append((times[i], times[i+1]))
                start_on = times[i+1]

        on_periods.append((start_on, times[-1]))

        if state == "On":
            onTime = "Fan ACTIVE from: "
            for each_time in on_periods:
                if each_time != on_periods[-1]:
                    onTime += f"{each_time[0]}ms to {each_time[1]}ms, "
                else:
                    onTime += f"{each_time[0]}ms to {each_time[1]}ms"
            return onTime
        else:
            offTime = "Fan OFF from: "
            for each_time in on_periods:
                if each_time != off_periods[-1]:
                    offTime += f"{each_time[0]}ms to {each_time[1]}ms, "
                else:
                    offTime += f"{each_time[0]}ms to {each_time[1]}ms"
            return offTime
```

```
    def Transitions(self):
        """
        Detects ON/OFF transitions.

        Returns:
            List[dict]: List of transition events.
        """
        times = [each_time.GetTime() for each_time in self.ValueReadings if
each_time.GetValue() > 300]


        interval = 200
        events = []
        previous_state = "OFF"   # assume fan starts OFF

        for i in range(len(times)-1):
            diff = times[i+1] - times[i]

            if diff == interval:
                current_state = "ON"
            else:
                current_state = "OFF"

            if current_state != previous_state:
                events.append({
                    "timestamp": f"{times[i]}ms",
                    "event": current_state
                })

            previous_state = current_state

        return events

# ---------ENERGY ESTIMATION---------
    def EstPower(self, voltage):
        """
        Estimates power using RMS current.

        Args:
            voltage (float): Supply voltage.

        Returns:
            float: Estimated power (W).
        """
```

```
        return (voltage * self.RMS()) * 0.8 #0.8 is the estimated Power Factor



    def EstEnergy(self, voltage):
        """
        Estimates energy consumption.

        Returns:
            float: Energy in kWh.
        """
        times = [each_time.GetTime() for each_time in self.ValueReadings if
each_time.GetValue() > 300]

        interval = 200

        on_periods = []

        start_on = times[0]

        for i in range(len(times)-1):
            diff = times[i+1] - times[i]

            if diff != interval:
                on_periods.append((start_on, times[i]))
                start_on = times[i+1]

        on_periods.append((start_on, times[-1]))

        total_on = 0
        for each_period in on_periods:
            total_on += each_period[1] - each_period[0]

        return (self.EstPower(voltage) * (total_on / 3600)) / 1000

    def EstEnergyVals(self, voltage):
        """
        Calculates estimated energy contribution for each reading.

        Args:
            voltage (float): Supply voltage.

        Returns:
            List[float]: Estimated energy values for each reading.
        """
```

```
        values = [each_value.GetValue()*voltage*0.8*(0.2/3600) for each_value
in self.ValueReadings]
        return values



# ---------REPORTING---------
    def BuildTimeline(self):
        """
        Builds time and state arrays for plotting.
        """
        self.time = np.arange(self.lowertime, self.uppertime-200,200)  # in ms

        self.state = []
        for each_value in self.ValueReadings:
            if each_value.GetTime() in self.time:
                if each_value.GetValue()>300:
                    self.state.append(1)
                else:
                    self.state.append(0)



    def CurrentVsTimePlot(self, unit):
        """
        Plots measured current against time.

        Args:
            unit (str): Time unit for the x-axis.
                Accepted values are "Seconds", "Minutes", or any other value
                for milliseconds.

        Returns:
            None
        """
        allowedValues = []
        for each_value in self.ValueReadings:
            if each_value.GetTime() >= self.lowertime and each_value.GetTime()
<= self.uppertime:
                allowedValues.append(each_value)

        Values = [each_entry.GetValue() for each_entry in allowedValues]

        Times = []
        if unit == "Seconds": #User wants graph plotted by seconds
            for each_value in allowedValues:
                Times.append(each_value.GetSeconds())
        elif unit == "Minutes": #User wants graph plotted by minutes
```

```
            for each_value in allowedValues:
                Times.append(each_value.GetMinutes())
        else: #User wants graph plotted by milliseconds
            for each_value in allowedValues:
                Times.append(each_value.GetTime())

        plt.figure(figsize=(10, 5))

        plt.plot(Times, Values)
        if unit == "Seconds":
            plt.xlabel("Time/seconds")
        elif unit == "Minutes":
            plt.xlabel("Time/minutes")
        else:
            plt.xlabel("Time/milliseconds")
        plt.ylabel("Current")
        plt.title("Current over time")

        plt.tight_layout()

        plt.show()


    def EnergyConsumptionPlot(self):
        """
        Plots estimated energy consumption values over time.

        Returns:
            None
        """
        plt.figure(figsize=(12, 5))
        plt.plot([each_time.GetTime() for each_time in self.ValueReadings],
self.EstEnergyVals(5))
        plt.xlabel("Time (ms)")
        plt.ylabel("Energy Consumption")
        plt.title("Energy Consumption Over Time")
        plt.show()


    def StateVsTimePlot(self):
        """
        Plots detected device state over time.

        The y-axis uses 1 for ON and 0 for OFF.
```

```
        Returns:
            None
        """
        plt.figure(figsize=(12, 5))
        plt.step(self.time, self.state, where='post')
        plt.xlabel("Time (ms)")
        plt.ylabel("State (On=1 / Off=0)")
        plt.title("Device State Over Time")
        plt.ylim(-0.1, 1.1)
        plt.show()


    def Summary(self):
        """
        Prints a summary of key metrics.
        """
        times = []
        for each_value in self.ValueReadings:
            if each_value.GetValue() >300: #Alter based on off threshold
                times.append(each_value.GetTime())

        interval = 200

        on_periods = []

        start_on = times[0]

        for i in range(len(times)-1):
            diff = times[i+1] - times[i]

            if diff != interval:
                on_periods.append((start_on, times[i]))
                start_on = times[i+1]

        on_periods.append((start_on, times[-1]))

        total_on = 0
        for each_period in on_periods:
            total_on += each_period[1] - each_period[0]

        print(f"""
Device: Fan
Runtime: {total_on/1000} seconds
RMS Current: {round(self.RMS(),1)} A
Peak Current: {self.Peak()} A
Mean Current: {self.Mean():.2f} A
```

```
Std Dev: {self.Stdev():.2f} A
Estimated Energy: {round(self.EstEnergy(5),2)} kWh
Duty Cycle: {round(self.DutyCycle(),1)}%
              """)

valueanalysis = ValueAnalysis(800, 9600, "test_energy_data_threshold300.csv")
valueanalysis.Summary()
valueanalysis.EnergyConsumptionPlot()
valueanalysis.StateVsTimePlot()
valueanalysis.CurrentVsTimePlot("Seconds")
```

5 Final Product

- Showcase the completed Mouse, highlighting its features and functionalities.
- Include images or diagrams to visually support the description.

6 Investigation of complex problem with laboratory skills - individual

7 Sustainability

7.1 Engineering for Circular Economy- individual

Which of the 4Rs were investigated and technologies used

This project primarily investigates **Reduce, Reuse, and Repair** within the design process. Reduction is achieved by minimising energy waste through real-time monitoring and analysis of appliance usage. Reuse is incorporated by using existing development hardware such as Arduino boards and modular sensors rather than designing a new dedicated PCB system. Repairability is supported through a modular architecture, where individual components such as the current sensor, microcontroller, or signal conditioning circuit can be replaced independently. Technologies used include embedded systems, analogue signal conditioning circuits, and Python-based data analysis to support efficient and sustainable system design. These approaches align with circular economy principles that prioritise resource efficiency and product longevity [1], [2].

Life Cycle Assessment

The project has a measurable environmental impact across its life cycle, including material extraction, manufacturing, use phase, and end-of-life disposal. The use phase provides the largest opportunity for CO_2 reduction, as the system enables identification of inefficient appliance behaviour and unnecessary runtime. By providing data-driven

insights, the system encourages energy-saving decisions, reducing electricity consumption and therefore lowering indirect CO_2 emissions associated with power generation [3], [4].

Assumptions:

Arduino + sensor + components mass ≈ 150 g electronics
Embedded electronics footprint ≈ 20–50 kg CO_2e per kg
Device lifetime: 3 years
Device consumption: ~0.5 W
Monitored appliance: 50 W fan
Behaviour improvement: 10% runtime reduction

These assumptions are based on typical lifecycle assessments of small electronic devices and average electricity carbon intensity values [3], [5].

CO2 footprint calculation

1. Raw materials + manufacturing
   0.15 kg × 30 kg CO_2/kg ≈ 4.5 kg CO_2
   (This aligns with typical embodied carbon values for electronic components [5])

2. **Use phase**
   0.5 W × 24 h × 365 ≈ 4.38 kWh/year
   Over 3 years: ≈ 13.1 kWh
   Using ~0.233 kg CO_2/kWh →
   = ~3.05 kg CO_2
   (This uses UK grid carbon intensity estimates [4])

3. Total system footprint
   ~7.5 kg CO_2 over lifecycle

Energy savings enabled by system

1. Fan example:
   50 W device × 4 h/day = 73 kWh/year
   10% reduction → 7.3 kWh saved/year
   Over 3 years: 21.9 kWh saved

2. CO_2 reduction:
   21.9 × 0.233 ≈ 5.1 kg CO_2 saved

Energy savings through behaviour change and monitoring are supported by energy efficiency studies showing that feedback systems can reduce consumption by 5–15% [3].

Final insight

1. System CO2 cost: 7.5kg
2. CO2 saved: 5.1kg

3. Net = -2.4kg CO2

This demonstrates that while the system has an embodied environmental cost, it can offset a significant portion of this through improved energy efficiency, especially when applied to multiple devices [3], [5].

Monetary value created

The system creates monetary value by reducing electricity consumption through improved user awareness of energy usage. By identifying inefficient usage patterns, such as leaving devices running unnecessarily, the system enables behavioural changes that lower electricity bills. Additionally, the use of low-cost, reusable components reduces initial development cost compared to commercial smart energy monitors.

Assume:

- Electricity cost: £0.30 per kWh
- Energy save: 21.9 kWh over 3 years

Savings:

- 21.9 * 0.30 = £6.57 saved per device

System cost:

- Arduino + sensor + components ~ £15 - £20

Lifcycle economic analysis

- Initial cost: ~£18
- Savings (single appliance): ~£6.57
- Net: −£11.43 (loss for single device)

If used across 3–4 appliances:

- Savings ≈ £20–£25
- Net positive value achieved

This reflects typical findings that energy monitoring systems become economically beneficial when applied at scale rather than to a single load [3].

7.2 Sustainable Materials Management- individual

critical raw material or a hazardous material (one material only per skill claim) within your project – so choose difrernet material to write about

Chosen material: Copper (critical raw material)

Where the material exists in the project

Copper is extensively used throughout the smart energy monitoring system, particularly in PCB traces, connecting wires, and internal circuitry within the current sensor and

microcontroller. It forms the primary conductive pathway for both analogue signal transmission and power distribution. Additionally, copper is present in connectors and solder joints, ensuring reliable electrical connections between components within the sensing and data acquisition subsystems. This widespread use of copper in electronic systems is well documented in electrical engineering literature and industry standards [1], [2].

Why it has been used

Copper is selected due to its excellent electrical conductivity (~5.96×10^7 S/m), low resistivity, and high reliability in electronic systems. These properties are essential for accurate current measurement, as signal degradation must be minimised to maintain measurement precision. Furthermore, copper exhibits high thermal conductivity, which assists in dissipating heat within circuits. Its compatibility with PCB manufacturing processes and cost-effectiveness make it the industry standard for electrical interconnections [1], [3].

Why it is a critical raw material

Copper is classified as a critical raw material due to its high demand in electrical, electronic, and renewable energy systems, combined with supply risks and environmental impacts associated with extraction. Mining and refining processes are energy-intensive and contribute significantly to greenhouse gas emissions. According to the European Commission, copper is strategically important for electrification and renewable energy technologies, while also being subject to supply chain constraints [4]. This makes its sustainable management essential in modern engineering design, particularly as demand increases with global energy transition efforts [5].

Engineering solution for sustainability

Several engineering strategies can be applied to improve the sustainability of copper use in this project:

- Material reduction: Optimising PCB layout to minimise unnecessary copper trace length and thickness
- Design for disassembly: Ensuring components can be easily separated to allow copper recovery at end-of-life
- Recycling integration: Using components manufactured with recycled copper where available
- System efficiency improvements: Reducing overall system energy waste indirectly reduces demand for copper-intensive energy infrastructure

These approaches align with circular economy principles that prioritise resource efficiency and material recovery [6].

Copper is difficult to replace due to its unmatched combination of conductivity, cost-effectiveness, and manufacturability. Therefore, improving lifecycle management rather than substitution is the most viable sustainability approach.

Justification

Copper plays a central role in both the electrical performance and environmental impact of the system. Its presence directly influences signal integrity, measurement accuracy, and energy efficiency. Because the project is fundamentally an electrical sensing and monitoring system, copper usage is unavoidable and highly significant. Its critical role in both functionality and sustainability demonstrates the importance of responsible material management in engineering design [1], [5].

7.3 Sustainable Energy Management

Life Cycle Assessment of the Smart Energy Monitoring Project

This life cycle assessment (LCA) evaluates the environmental impacts and energy performance of the smart energy monitoring system across all stages: raw material extraction, processing and manufacturing, assembly, use, maintenance, and end-of-life. The assessment also identifies opportunities to improve energy performance through energy efficiency, conservation, and data-driven decision-making. Lifecycle-based approaches are widely used to evaluate environmental impact and optimise engineering systems [1].

Raw Material Extraction

The project relies on materials such as copper, silicon, plastics, and small quantities of rare materials used in semiconductor devices. The extraction of these materials is energy-intensive and contributes to environmental degradation, including emissions and resource depletion. These impacts are well documented in lifecycle studies of electronic devices [2]. The environmental impact is reduced by using existing lab equipment and commonly available components, avoiding the need for new manufacturing. Material efficiency is further improved through compact system design with minimal additional hardware.

Processing and Manufacturing

Manufacturing processes such as semiconductor fabrication and PCB production have high energy demands and associated emissions. These stages often dominate the embodied energy of electronic systems [2]. In this project, environmental impact is reduced by using pre-manufactured modules such as Arduino boards and sensors rather than producing custom electronics. The use of breadboards and reusable prototyping components avoids additional fabrication, reducing both material consumption and manufacturing energy use.

Assembly and Integration

The assembly phase involves wiring, signal conditioning circuit construction, and microcontroller integration. While relatively low impact, inefficient design iterations can increase material and energy use. Energy performance is improved by adopting a modular design approach, allowing subsystems to be tested independently. This reduces repeated assembly processes and minimises unnecessary energy consumption, aligning with efficient engineering practices [3].

Use Phase

The use phase is the most significant stage in terms of energy impact, as the system operates continuously to monitor electrical current and analyse appliance behaviour. The monitoring system itself consumes minimal power (low-power microcontroller and sensor), but its primary benefit lies in enabling energy savings in external devices.

Energy performance improvements

- Identifying unnecessary appliance runtime
- Detecting idle or standby energy consumption
- Providing data for optimised usage patterns
-

Studies show that feedback-based energy monitoring systems can reduce energy consumption by 5–15% through behavioural change [4].

The use of Python-based data processing enables automated analysis, reducing manual intervention and supporting continuous optimisation. Energy conservation is achieved by operating the system efficiently and minimising idle processing.

Quantified Energy Performance Improvements

The smart energy monitoring system improves energy performance by identifying unnecessary appliance runtime and encouraging reduced usage. For this project, a 50W appliance such as a small fan was used as the monitored load.
Baseline energy use:

- 50 W × 4 hours/day = 200 Wh/day = 0.2 kWh/day

Annual baseline:

- 0.2 kWh/day × 365 = 73 kWh/year

If the monitoring system helps reduce unnecessary runtime by 10%, the improved annual energy use becomes:

- 73 kWh/year × 0.90 = 65.7 kWh/year

Energy saving:

- 73 − 65.7 = 7.3 kWh/year

Using an estimated UK grid carbon factor of 0.233 kg CO_2/kWh, this gives a carbon saving of:

- 7.3 × 0.233 = 1.70 kg CO_2/year

Over a 3-year project lifecycle:

- 7.3 × 3 = 21.9 kWh saved
- 1.70 × 3 = 5.1 kg CO_2 saved

These values align with reported energy savings from monitoring-based interventions [4].

Maintenance

Maintenance involves recalibration, software updates, and occasional component replacement. The modular design allows individual components to be replaced without discarding the entire system, reducing waste. Efficient diagnostics using software tools reduce repeated physical testing, lowering both energy and material consumption over

time. This reflects lifecycle optimisation strategies commonly used in sustainable engineering [1].

End-of-Life

At the end of its lifecycle, components such as the Arduino and sensors can be reused. Electronics should be directed to appropriate recycling streams to recover valuable materials such as copper and prevent environmental contamination. Design for disassembly improves recyclability and reduces landfill waste, supporting circular economy principles [6].

Conclusion

The LCA highlights that the primary environmental impacts arise from raw material extraction and the indirect effects of energy use in monitored devices. By focusing on low-power system design, modular construction, reuse of components, and enabling energy-efficient behaviour in external appliances, the project significantly reduces its overall environmental footprint. The system demonstrates how relatively simple monitoring technology can contribute to improved energy management and more sustainable engineering practices.

Reflection of relevance

Sustainable energy management is directly relevant to this project because the system is designed to monitor, analyse, and reduce unnecessary energy use. Without this, the system would only collect data without delivering environmental benefit. By identifying waste, reducing runtime, and quantifying savings, the system becomes a decision-support tool. This demonstrates how embedded systems and data analysis can contribute to real-world energy conservation, aligning with global sustainability goals [3], [4].

7.4 Regenerative Design- individual

Whole System Considered

The system is defined using a multi-level systems thinking approach, considering interactions across four scales: site, locality, city/region, and ecosystem/global. Systems thinking is widely used in sustainable engineering to analyse complex interactions rather than isolated components [1].

- Site: The smart energy monitoring device (current sensor, signal conditioning circuit, Arduino, and Python analysis system) and the monitored appliance
- Locality: The building or home where the system is deployed, including user behaviour and energy consumption patterns
- City/Region: The electrical grid supplying energy and the aggregated impact of reduced energy demand
- Ecosystem/Global: Environmental systems affected by energy production, including emissions, resource extraction, and climate change

This holistic definition reflects regenerative design principles, where systems are interconnected and influence each other across multiple scales [2].

Positive Environmental Impact and Biomimicry (Phases 1–4)

The regenerative design approach is analysed using phases based on consumption, pressures, habitat, and societal benefit. These phases align with ecological frameworks used to understand human-environment interactions [3].

Phase 1 – Consumption Drivers:
The system reduces energy consumption by identifying inefficient appliance use. This mimics natural ecosystems where energy flows are optimised and waste is minimised. Efficient energy use is a key characteristic of biological systems [4].

Phase 2 – Human Pressures:
By reducing electricity demand, the system lowers indirect pollution from power generation. This reflects natural regulatory systems such as carbon cycles, which balance inputs and outputs to maintain environmental stability [3].

Phase 3 – Habitat Status:
Reduced energy demand decreases the need for expansion of energy infrastructure, indirectly preserving ecosystems. This aligns with regenerative design principles that aim to minimise environmental disturbance and maintain system balance [2].

Phase 4 – Nature's Contribution to People:
The system improves human well-being by reducing energy costs and increasing awareness of sustainable behaviour. This mirrors natural systems where efficient resource use supports long-term resilience and survival [4].

Overall, the system demonstrates **biomimicry** by emulating efficiency, feedback, and balance found in natural ecosystems, rather than simply reducing environmental harm [4].

Interdisciplinary Collaboration

The project requires collaboration across multiple disciplines. Electrical and electronic engineering is central for sensing and circuit design. Computer science supports data processing and automation. Energy engineering is required to interpret consumption data and optimise efficiency. Environmental engineering contributes lifecycle assessment and sustainability evaluation. Behavioural science helps understand user responses to feedback, while economics informs cost-benefit analysis and system adoption.

Such interdisciplinary collaboration is essential in sustainable system design, where technical, environmental, and social factors must be integrated [1].

Long-Term Perspective and Positive Feedback Loop

The system incorporates a positive feedback loop, where energy monitoring leads to behavioural change, which reduces energy consumption and reinforces continued

system use. Feedback loops are a key feature of both ecological systems and regenerative design, enabling continuous adaptation and improvement [2], [4].

For example, as users observe energy consumption data, they reduce unnecessary appliance usage. This leads to lower energy demand and environmental impact, reinforcing the value of the monitoring system.

Over time, the system can evolve by integrating advanced analytics, automation, or machine learning, allowing adaptation to changing behaviour and energy systems. This reflects the adaptive nature of living systems, which continuously respond to environmental changes [4].

The modular hardware design also supports long-term adaptability, enabling upgrades or replacement without full system redesign. At a larger scale, widespread adoption could influence grid-level demand, supporting more sustainable energy systems.

This demonstrates a regenerative approach where the system not only reduces impact but continuously improves environmental performance through self-reinforcing feedback mechanisms [2].

Phase / Level	Site (Device)	Locality (Building)	City/Region	Global
Phase 1: Consumption	Low-power sensing system reduces device-level energy use	Identifies inefficient appliance usage	Reduced aggregate electricity demand	Lower resource extraction and energy production
Phase 2: Pressures	Minimal additional energy use from monitoring system	Reduced household energy waste	Lower emissions from power plants	Reduced global CO_2 emissions
Phase 3: Habitat	Small-scale system with minimal physical footprint	Less need for additional energy infrastructure	Reduced strain on grid expansion	Lower ecosystem disruption from energy production
Phase 4: Contribution	Provides real-time data feedback	Improves user awareness and behaviour	Supports demand-side energy management	Contributes to sustainability goals
Phase 5: Time/Feedback	System logs and learns from data	Users adapt behaviour over time	Demand patterns evolve with adoption	Long-term reduction in environmental impact

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	Degenerative characteristics	always -100	usually -75	sometimes -50	slight -25	balanced 0	slight 25	sometimes. 50	usually. 75	always. 100	Regenerative characteristics	Value weighting Recommended range 1 to 5	Weighted outcome
Contextual factors	Does not provide actionable energy insights					100					Enables real-time energy awareness and decision-maki	4	400
	Encourages inefficient energy behaviour					100					Promotes energy-efficient behaviour through feedbac	4	400
	Ignores local energy consumption patterns					50					Adapts to local usage patterns	2	100
	No connection to user behaviour					100					Integrates user interaction and feedback	5	500
	Data not accessible or understandable					75					Provides clear and usable data insights	3	225
	Works in isolation from environment					50					Integrates into broader energy system	5	250
	Does not support behavioural change					100					Encourages sustainable habits	5	500
	No consideration of energy waste					100					Identifies and reduces wasted energy	4	400
	Static system with no adaptation					50					Dynamically adapts to usage patterns	3	150
	Depends on grid electricity				-25						Does not depend on grid electricity	3	-75
Project factors	Uses unnecessary materials					25					Minimises material use with compact design	3	75
	Requires constant replacement					50					Designed for long lifespan and reuse	3	150
	Uses non-recyclable components					25					Uses recyclable/modular components	4	100
	High power consumption					75					Low-power embedded system	5	375
	No use of renewable energy context					75					Supports renewable-aware energy decisions	4	300
	Requires frequent maintenance					50					Minimal maintenance due to simple design	3	150
	Single-use design					50					Designed for reuse across multiple systems	4	200
	Generates no system-level benefit					75					Enables wider energy system optimisation	5	375
	No scalability					50					Scalable across multiple devices/users	4	200
	No environmental feedback					100					Provides continuous environmental feedback fo	5	500
												Project Score	5275
	Regenerative Checklist											Regenerative Maximum	7800
												Project performance	68%
												Your project is	Highly regenerative

8 Professional Skills

8.1 Risk Mitigation

8.2 Change Management- individual

8.3 Feasibility Study

8.4 Quality Management System

8.5 Standards and Regulations - individual

The IEC 61010-1, published by the International Electrotechnical Commission, was applied to ensure safe design of the current sensing and measurement circuitry in the smart energy monitoring system [1].

IEC 61010-1 defines safety requirements for electrical measurement equipment, including protection against electric shock, insulation coordination, and safe circuit design. This standard is directly applicable to the project as it involves measuring current in electrical systems using sensors and microcontrollers. The project interfaces with powered devices, which introduces potential electrical hazards. Compliance ensures safe operation during testing and use, particularly when monitoring real appliances. Applying such standards is essential for professional engineering practice and risk mitigation [1], [2].

IEC 61010-1 significantly influenced the design of the current sensing and signal conditioning stages of the system. To comply with safety requirements, the design avoided direct electrical connection to high-voltage circuits and instead utilised an isolated current sensor module. This ensured galvanic isolation between the measurement system and the electrical load, reducing the risk of electric shock. Additionally, component selection and circuit layout were guided by safety considerations such as limiting current levels, ensuring adequate spacing between conductive tracks, and preventing overheating. Low-voltage testing procedures were implemented before connecting to higher-power devices, aligning with recommended safe testing practices.
These decisions demonstrate how the standard influenced both hardware architecture and testing methodology. By incorporating safety principles early in the design process, the system not only meets functional requirements but also aligns with recognised engineering best practices. This reflects the role of standards in improving reliability, safety, and professional accountability in engineering design [1], [3].

Future standards could expand beyond safety to include sustainability requirements for electronic systems. While existing standards such as IEC 61010-1 focus on safety and performance, there is increasing need for regulations addressing lifecycle environmental impact. New standards could mandate energy efficiency thresholds, design for disassembly, and the use of recyclable materials. Additionally, integrating sustainability metrics such as carbon footprint reporting into standards would encourage engineers to consider environmental impacts during design. This aligns with global efforts to reduce emissions and improve sustainable engineering practices [4], [5].

8.6 Time Management

8.7 Project Management

9 Conclusion

- Summarize the achievements of the project.
- Reflect on the skills and knowledge gained.
- Conclude with remarks on the project's contribution to personal and educational growth.

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10 References (including Quality Referencing) - individual

Put each main text into gpt to reword with difference references so guarantees similar but different information and add your references here like mine

Being security minded

[1] ISO, ISO/IEC 27001: Information security management systems—Requirements, 2022. https://www.iso.org/standard/82875.html
Strong reference: This is a current and authoritative international standard for information security management. It is highly relevant to the project because it supports data protection, access control, and risk management.

[2] NIST, Framework for Improving Critical Infrastructure Cybersecurity, 2018. https://www.nist.gov/cyberframework
Strong reference: This is a widely used cybersecurity framework. It is relevant because it supports a structured risk-based approach to identifying, protecting, detecting, responding to, and recovering from security risks.

[3] ENISA, Guidelines for Securing the Internet of Things, 2020. https://www.enisa.europa.eu/publications
Strong reference: This is directly relevant to embedded and IoT-style monitoring systems. It is reasonably recent and useful for understanding device security, data protection, and connected-system risks.

[4] A. R. Sadeghi, C. Wachsmann, and M. Waidner, "Security and privacy challenges in industrial Internet of Things," Proc. Design Automation Conf., pp. 1–6, 2015. https://doi.org/10.1145/2744769.2747942
Strong reference: This peer-reviewed paper is relevant because it explains security and privacy risks in connected embedded systems. Although older, it remains useful for supporting the technical privacy argument.

Circular economy

[1] A. Tukker, "Product services for a resource-efficient and circular economy," Journal of Cleaner Production, vol. 97, pp. 76–91, 2015. https://doi.org/10.1016/j.jclepro.2013.11.049
Strong reference: Peer-reviewed journal article providing strong theoretical support for circular economy principles such as reduce, reuse, and repair. Widely cited and highly relevant, though slightly older.

[2] European Commission, Circular Economy Action Plan, 2020. https://eur-lex.europa.eu
Strong reference: Authoritative policy document directly supporting circular design strategies. Highly relevant and relatively recent, though less technical than journal sources.

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[3] International Energy Agency, Energy Efficiency 2023, 2023. https://www.iea.org/reports/energy-efficiency-2023
Strong reference: Recent and highly authoritative report. Provides strong evidence for energy savings through monitoring and behavioural change, making it directly relevant to the project.

[4] UK Government Department for Business Energy & Industrial Strategy, UK Government GHG Conversion Factors, 2023. https://www.gov.uk/government/collections/government-conversion-factors-for-company-reporting
Strong reference: Official government data source providing accurate and current carbon intensity values. Highly reliable and directly supports CO_2 calculations.

[5] L. Belkhir and A. Elmeligi, "Assessing ICT global emissions footprint," Journal of Cleaner Production, vol. 177, pp. 448–463, 2018. https://doi.org/10.1016/j.jclepro.2017.12.239
Strong reference: Peer-reviewed study quantifying carbon footprint of electronic devices. Highly relevant for embodied emissions assumptions, though slightly dated.

Materials Management

[1] W. H. Hayt and J. A. Buck, Engineering Electromagnetics, 8th ed. New York, NY, USA: McGraw-Hill, 2012.
Strong reference: This is a well-established textbook providing authoritative data on electrical conductivity and material properties. It is highly reliable and relevant, though not the most recent source.

[2] Copper Development Association, Copper in Electrical Applications, 2022. https://copperalliance.org
Strong reference: Industry-backed source directly focused on copper usage in electronics. Highly relevant and practical, though less academically rigorous than peer-reviewed literature.

[3] M. M. Schwartz, Encyclopedia of Materials, Parts and Finishes, 2nd ed. Boca Raton, FL, USA: CRC Press, 2002.
Moderate reference: Provides useful background on material properties and manufacturing processes. However, it is older, reducing timeliness, though still technically valid.

[4] European Commission, Critical Raw Materials Resilience: Charting a Path towards greater Security and Sustainability, 2020. https://eur-lex.europa.eu
Strong reference: Highly authoritative policy document defining critical raw materials. Directly relevant to copper classification and sustainability considerations. Reasonably recent and widely used.

[5] International Energy Agency, The Role of Critical Minerals in Clean Energy Transitions, 2021. https://www.iea.org/reports

Strong reference: Provides strong evidence linking copper demand to energy transition technologies. Highly relevant and relatively recent, with strong technical and policy credibility.

[6] A. Tukker, "Product services for a resource-efficient and circular economy," Journal of Cleaner Production, vol. 97, pp. 76–91, 2015.
https://doi.org/10.1016/j.jclepro.2013.11.049
Strong reference: Peer-reviewed journal article supporting circular economy strategies. Highly relevant for sustainability solutions, though slightly older.

Energy management

[1] ISO, ISO 14040: Environmental management—Life cycle assessment—Principles and framework, 2006. https://www.iso.org
Strong reference: Foundational standard for lifecycle assessment methodology. Highly relevant and widely accepted, though older, but still the industry benchmark.

[2] L. Belkhir and A. Elmeligi, "Assessing ICT global emissions footprint," Journal of Cleaner Production, vol. 177, pp. 448–463, 2018.
https://doi.org/10.1016/j.jclepro.2017.12.239
Strong reference: Peer-reviewed study quantifying environmental impact of electronics. Highly relevant to material and manufacturing stages, though moderately dated.

[3] International Energy Agency, Energy Efficiency 2023, 2023.
https://www.iea.org/reports/energy-efficiency-2023
Strong reference: Recent and authoritative report. Provides strong evidence for energy efficiency improvements and relevance of monitoring systems.

[4] S. Darby, "The effectiveness of feedback on energy consumption," Environmental Change Institute, University of Oxford, 2006. https://www.eci.ox.ac.uk
Strong reference: Widely cited study on energy feedback systems. Although older, it is foundational and highly relevant to behavioural energy savings.

[5] UK Government Department for Business Energy & Industrial Strategy, GHG Conversion Factors, 2023. https://www.gov.uk
Strong reference: Official and current dataset providing accurate carbon intensity values. Highly reliable and directly supports CO_2 calculations.

[6] A. Tukker, "Product services for a resource-efficient and circular economy," Journal of Cleaner Production, vol. 97, pp. 76–91, 2015.
https://doi.org/10.1016/j.jclepro.2013.11.049
Strong reference: Provides strong support for circular economy and lifecycle sustainability strategies. Highly relevant, though slightly older.

Regenerative design

[1] P. Senge, The Fifth Discipline: The Art and Practice of the Learning Organization, New York, NY, USA: Doubleday, 2006.

Strong reference: Foundational text on systems thinking. Highly relevant for understanding interconnected systems and interdisciplinary design, though not specifically focused on engineering.

[2] J. T. Lyle, Regenerative Design for Sustainable Development, New York, NY, USA: Wiley, 1994.
Strong reference: Key text defining regenerative design principles. Highly relevant to the project's approach, though older; still widely cited in sustainability literature.

[3] Millennium Ecosystem Assessment, Ecosystems and Human Well-being, 2005. https://www.millenniumassessment.org
Strong reference: Authoritative global assessment linking ecosystems and human activity. Provides strong theoretical grounding for environmental phases, though older.

[4] J. Benyus, Biomimicry: Innovation Inspired by Nature, New York, NY, USA: HarperCollins, 2002.
Strong reference: Widely recognised source on biomimicry principles. Highly relevant to explaining natural system analogies, though not recent.

[5] International Energy Agency, Energy Efficiency 2023, 2023. https://www.iea.org/reports/energy-efficiency-2023
Strong reference: Recent and authoritative report supporting energy reduction impacts and system-level benefits. Highly relevant and up-to-date.

Standards and regulations

[1] IEC, IEC 61010-1: Safety requirements for electrical equipment for measurement, control, and laboratory use, 3rd ed., 2010. https://www.iec.ch
Strong reference: This is a primary international safety standard directly relevant to electrical measurement systems. It is authoritative and highly credible, though slightly dated. Its continued use in industry ensures strong relevance despite not being the most recent revision.

[2] ISO, ISO/IEC Guide 51: Safety aspects—Guidelines for their inclusion in standards, 2014. https://www.iso.org/standard/53940.html
Strong reference: This provides foundational guidance on safety principles across engineering standards. While not specific to the project, it is highly relevant for understanding why safety standards exist. It is authoritative but somewhat general and moderately dated.

[3] IEEE, "IEEE Code of Ethics," 2020. https://www.ieee.org/about/corporate/governance/p7-8.html
Strong reference: This is a widely recognised professional standard outlining ethical engineering responsibilities. It is relevant for justifying safety-conscious design decisions. While not technical, it strengthens professional context and remains current and authoritative.

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[4] IEA, Energy Efficiency 2023, 2023. https://www.iea.org/reports/energy-efficiency-2023
Strong reference: This is a recent and highly authoritative report providing up-to-date data on energy efficiency and sustainability trends. It is directly relevant to the project's impact on energy use and strengthens the sustainability argument significantly.

[5] A. Tukker, "Product services for a resource-efficient and circular economy," Journal of Cleaner Production, vol. 97, pp. 76–91, 2015. https://doi.org/10.1016/j.jclepro.2013.11.049
Strong reference: This is a peer-reviewed journal article providing strong theoretical grounding in sustainability and circular economy principles. While slightly older, it remains widely cited and highly relevant for supporting sustainability arguments in engineering design.

Appendices

[A section break has been added before the Appendices to allow the page to be in landscape format for the appendices only]

Appendix A: Design and Development Matrix

See the instructions in the lab script for writing the risk mitigation section.

Needs analysis		Target specifications			Technical solution	
Ref Need	Description	Ref Spec	Description	Objectives measurable + tolerance	Possible solutions	The chosen one

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Appendix B: Risk Assessment & Mitigation

Severity (a)	Likelihood of Occurrence (b)
1. – Trivial (No physical injury, no component damage, project timeline unaffected) 2. – Minor (Minor pinch from fan blade, single cheap component destroyed like a resistor, <1 hour delay) 3. – Moderate (First aid required, expensive component destroyed like the microcontroller, 1-2 day delay) 4. – Serious (Hospital visit required, total system destruction, project failure likely) 5. – Catastrophic (Permanent severe injury or death)	1 – Remote (1% - 10% chance of occurring) 2 – Unlikely (11% - 30% chance of occurring) 3 – Possible (31% - 50% chance of occurring) 4 – Likely (51% - 80% chance of occurring) 5 – Very likely (>80% chance of occurring)

Risk Assessment Matrix

(B)↓ (A)→	Trivial	Minor	Moderate	Serious	Catastrophic
Remote	1	2	3	4	5
Unlikely	2	4	6	8	10
Possible	3	6	9	12	15
Likely	4	8	12	16	20
Very likely	5	10	15	20	25

Risk Rating Bands (A x B)

LOW RISK	MEDIUM RISK	HIGH RISK
Continue, but review periodically to ensure controls remain effective	Continue, but implement additional reasonably practicable controls where possible and monitor regularly	STOP THE ACTIVITY Identify new controls. Activity must not proceed until risks are reduced to a low or medium level

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#	Hazard(s) identified	Existing controls & measures	Severity (a)	Likelihood (b)	Risk Rating (a x b)	Additional control/action required	Severity (a)	Likelihood (b)	Risk Rating (a x b)
1	Fast-spinning, exposed metal fan blades present a severe mechanical hazard, potentially causing deep lacerations to fingers if accidentally touched during active testing.	• Basic user awareness. • Keeping hands clear during operation.	4	4	16	• Install a rigid physical mesh guard over the plastic enclosure. • Enforce a strict "hands-off" policy while the system is powered to completely eliminate contact risk with the metal blades.	4	1	4
2	Stray conductive debris (e.g., metal shavings) falling directly across the exposed motor terminals or PCB traces, causing a localized short circuit that destroys the microcontroller.	• Visual inspection before use. • Keeping the lab bench clean.	4	3	12	• Apply a protective conformal coating across all exposed PCB traces. • Wrap the direct motor terminals in tightly fitted heat-shrink tubing to physically isolate the remaining bare electrical contacts.	4	1	4
3	Continuous high-current draw through the motor causing localized overheating, potentially softening or melting the plastic housing and compromising structural integrity.	• Operating in a well-ventilated lab. • Manual temperature checks.	3	3	9	• Upgrade to an appropriately rated motor driver with a heatsink. • Strategically design ventilation slits into the plastic housing to allow the fan's own airflow to provide active, continuous cooling.	3	1	3

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#	Hazard(s) identified	Existing controls & measures	Severity (a)	Likelihood (b)	Risk Rating (a x b)	Additional control/action required	Severity (a)	Likelihood (b)	Risk Rating (a x b)
4	A piece of debris jamming the 3 plastic gears, causing a motor stall and thermal runaway that could ignite the plastic housing and cause a severe lab fire.	• Manual observation. • Powering off if burning plastic smells occur.	5	3	15	• Implement an in-line fast-blow fuse to physically break the circuit during high-current stall conditions. • Integrate a thermistor to trigger an automatic shutdown before the plastic housing reaches its melting point.	2	1	2
5	High torque from the heavy metal fan blade causing the plastic gears to strip or shatter at maximum RPM, ejecting sharp plastic shrapnel that could cause serious eye injuries.	• Wearing standard lab safety glasses. • Keeping a safe distance.	4	3	12	• Replace the standard plastic gears with reinforced nylon alternatives to handle the metal blade's torque. • Thicken the plastic housing walls around the gearbox to safely contain any potential gear shrapnel.	1	1	1
6	Heavy rotational mass from the metal blade causing intense continuous vibration, gradually loosening structural screws and leading to sudden mechanical collapse.	• Standard screw tightening. • Periodic visual inspection of the housing.	3	4	12	• Apply thread-locking fluid to all structural screws connecting the motor to the plastic housing. • Add rubber dampening pads under the motor mount to absorb the metal blade's rotational vibrations.	3	1	3

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#	Hazard(s) identified	Existing controls & measures	Severity (a)	Likelihood (b)	Risk Rating (a x b)	Additional control/action required	Severity (a)	Likelihood (b)	Risk Rating (a x b)
7	The temperature sensor providing inaccurate data, causing the fan to fail to trigger, which allows ambient heat to build up inside the plastic housing.	• Basic sensor calibration. • Using digital over analog sensors.	3	3	9	• Implement software-side moving average filters to smooth noisy data. • Add a fail-safe threshold in Python that triggers maximum cooling if sensor readings abruptly drop to zero.	3	1	3
8	Incorrectly configuring the lab DC bench power supply (e.g., applying 30V instead of 12V), causing an immediate overvoltage event that destroys all connected control circuitry.	• Double-checking dials before plugging in. • Using standard lab protocols.	4	3	12	• Implement a Zener diode-based overvoltage protection circuit at the main power input. • Enforce a strict two-person visual verification of the DC supply configuration before physically connecting the system.	2	1	2